Generational Depth Estimation in Recursive Synthetic Text: From Discrete Boundaries to Continuous Decay

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Abstract

Large Language Model (LLM) generated text increasingly contaminates web corpora through recursive cycles: synthetic outputs become training inputs for subsequent generation. Detecting this contamination requires estimating the *generational depth* of text, yet existing detectors treat the problem as binary (human vs. machine). We formalize recursive generation as an iterated lossy channel and define K^*, the discrimination boundary beyond which consecutive depths become statistically indistinguishable. Controlled experiments across eight model configurations (five architecture families, 1B–8B parameters), three source domains, and four decoding strategies yield $K^* \in \{1, 2, 3\}$ with domain-dependent structure. However, two validation studies expose fundamental fragility in this discrete characterization. A naturalistic evaluation on Common Crawl data reveals severe lab-to-wild transfer degradation: depth–quality correlations are negligible (Spearman $\rho = -0.08$, $p = 0.07$ for AlignScore), and calibration collapses (ECE = 0.31). A subsampling power analysis shows that the canonical $K^* = 2$ finding on web text requires $n \geq 4{,}000$ samples per depth pair for 80% detection power, exposing a vanishingly small effect size (Cohen's $h \approx 0.20$). These results suggest that K^* as a discrete operational quantity is an artifact of experimental design choices rather than a robust property of recursive generation, motivating a shift toward continuous information decay characterization.

1 Introduction

Web-scale text corpora are increasingly contaminated by recursively generated synthetic content. When LLM outputs re-enter the training pipeline, the resulting text undergoes iterative quality degradation that binary detectors (human vs. machine) cannot capture. Understanding how many recursive passes a document has undergone, its *generational depth*, is critical for principled data curation.

Recent work on model collapse [Shumailov et al., 2024, Alemohammad et al., 2024, Dohmatob et al., 2024] demonstrates that recursive training on synthetic data leads to distributional narrowing. Binary detection methods such as DetectGPT, DivEye, and Min-K%++ distinguish human from machine text but do not estimate depth. Data selection approaches (DSIR, D4, DoReMi) perform distribution matching without depth awareness. None of these methods quantify the point at which consecutive generational depths become indistinguishable.

We formalize recursive generation as an iterated lossy channel and, via the Data Processing Inequality (DPI), prove that a finite discrimination boundary K^* must exist. We then measure K^* empirically across a comprehensive experimental matrix: eight model configurations spanning five architecture families (Pythia-1.4B, OLMo-1B, Qwen-1.5B, Qwen-7B, LLaMA-3.1-8B, Mistral-7B), three source domains (C4 web text, Wikipedia, arXiv), and four decoding strategies. The controlled experiments yield $K^* \in \{1, 2, 3\}$ with a domain-dependent hierarchy: Wikipedia saturates at $K^* = 1$, C4 web text at $K^* = 2$ (7/8 configurations), and arXiv at $K^* = 3$ (5/5 domain-competent models).

However, two additional validation experiments challenge this clean picture. First, a naturalistic evaluation on Common Crawl (exp-014) reveals that depth estimators trained under controlled conditions fail to generalize to wild data, with calibration error reaching ECE = 0.31

and depth–quality correlations falling to negligible levels. Second, a subsampling power analysis (exp-028) demonstrates that $K^* = 2$ on C4 requires $n \geq 4{,}000$ samples for reliable detection, exposing an effect size so small that its practical significance is questionable.

Together, these findings suggest that K^* as a discrete operational quantity may be an artifact of specific experimental configurations (sample size, threshold θ, domain) rather than a stable property of the recursive generation process. We argue that the field should move beyond discrete K^* toward characterizing continuous information decay curves.

Our contributions are:

- We formalize the discrimination boundary K^* via DPI theory and provide the first systematic empirical measurement across 8 model configurations, 3 domains, and 4 decoding strategies, finding $K^* \in \{1, 2, 3\}$ with domain-dependent structure.
- We expose the fragility of discrete K^* through two validation experiments: (a) lab-to-wild transfer failure on Common Crawl (ECE = 0.31, negligible depth–quality correlation), and (b) sample-size dependence of K^* determination ($n \geq 4{,}000$ required for 80% power).
- We argue for a paradigm shift from discrete K^* characterization to continuous information decay curves, grounded in the empirical evidence of K^*'s instability.

2 Method

2.1 Recursive Generation as Iterated Lossy Channel

Let X_0 denote human-authored text drawn from source distribution $\mathcal{D}$. Recursive generation produces a Markov chain $X_0 \to X_1 \to X_2 \to \cdots$, where $X_{k+1} = G(X_k)$ for a generation operator G (prompt-chain continuation or instruction-following rewrite). By the Data Processing Inequality, for any statistical functional T:

$$I(X_0; T(X_k)) \leq I(X_0; T(X_{k-1})) \leq \cdots \leq I(X_0; T(X_0)), \tag{1}$$

where $I(\cdot;\cdot)$ denotes mutual information. This monotonic decay guarantees the existence of a finite K^*_{theory} beyond which no statistical test can distinguish X_k from X_{k+1}.

2.2 Empirical Discrimination Boundary K^*

We define the empirical discrimination boundary $K^*_{\text{empirical}}$ as:

$$K^* = \max\{k : \text{AUC}(k \text{ vs } k{+}1) > \theta\}, \tag{2}$$

where AUC(k vs k+1) is the area under the ROC curve for a binary classifier distinguishing depth-k from depth-(k+1) text, and θ is a discrimination threshold. We set $\theta = 0.60$, corresponding to Cohen's $h \approx 0.20$ (small effect size), and require the lower bound of a bootstrap 95% confidence interval to exceed θ.

2.3 Feature Representation

We use a 15-dimensional model-free feature vector comprising: (i) token-level surprisal statistics (mean, variance, skewness, kurtosis) computed via a reference LM, (ii) vocabulary richness metrics (type-token ratio, hapax legomena ratio, Yule's K), (iii) syntactic complexity measures (mean dependency length, clause depth), (iv) repetition indicators (n-gram overlap ratios for $n \in \{2, 3, 4\}$), and (v) coherence features (entity grid transition probabilities, discourse connective density). These features are model-agnostic by design, enabling cross-model transfer.

Table 1: K^* determination on C4 web text ($\theta = 0.60$, continuation mode). AUC and 95% bootstrap CI for the boundary pair (depth 1 vs 2). 7/8 configurations yield $K^* = 2$.

Model	AUC 1v2	95% CI	K^*
Pythia-1.4B	0.6539	[0.645, 0.663]	2
Qwen-1.5B base	0.6262	[0.617, 0.635]	2
Qwen-7B	0.6178	[0.608, 0.626]	2
OLMo-1B	0.6145	[0.605, 0.625]	2
LLaMA-3.1-8B	0.6123	[0.603, 0.622]	2
Qwen-1.5B-Inst cont	0.6112	[0.602, 0.621]	2
Qwen-1.5B-Inst rw	0.6957	[0.686, 0.704]	2
Mistral-7B	0.5935	[0.584, 0.603]	1

Table 2: Domain-dependent K^* hierarchy ($\theta = 0.60$). Wikipedia saturates at $K^* = 1$, C4 at $K^* = 2$, arXiv at $K^* = 3$.

Domain	K^* (dominant)	Models agreeing
Wikipedia	1	6/6 (scorer-controlled)
C4 web text	2	7/8
arXiv	3	5/5 (domain-competent)

2.4 Experimental Matrix

The controlled experiments span:

- **Models**: Pythia-1.4B, OLMo-1B, Qwen-1.5B (base and instruct), Qwen-7B, LLaMA-3.1-8B, Mistral-7B
- **Domains**: C4 web text, Wikipedia, arXiv
- **Decoding**: greedy, nucleus ($p = 0.9$), temperature (0.7, 1.2)
- **Generation mode**: continuation (canonical) and instruction-following rewrite (ablation)

Each configuration generates depths 0 through 5, with $n = 5{,}000$ samples per depth in the primary C4 experiments.

3 Controlled Experiments: K^* Characterization

Under controlled lab conditions, the 15D feature classifier yields consistent K^* values across the experimental matrix. Table 1 summarizes the C4 web text results, where 7/8 configurations yield $K^* = 2$. Table 2 shows the domain hierarchy.

These controlled results present a clean narrative: K^* is low-bounded (≤ 3) and domain-dependent. The following two sections examine whether this narrative survives validation.

4 Naturalistic Common Crawl Validation (exp-014)

4.1 Setup

To evaluate whether the lab-trained depth estimator generalizes beyond controlled conditions, we applied it to $n = 10{,}000$ documents sampled from the Common Crawl 2025-51 snapshot. We computed predicted depth for each document and measured correlations with three independent quality proxies: AlignScore (factual consistency), vocabulary richness (type-token ratio, TTR), and perplexity (GPT-2).

Table 3: Correlation between predicted depth and quality metrics on Common Crawl 2025-51 ($n = 10{,}000$).

Metric Pair	Spearman ρ	p-value	95% CI
Pred Depth × AlignScore	−0.08	0.07	[−0.16, 0.01]
Pred Depth × Vocab Richness	−0.14	0.002	[−0.22, −0.05]
Pred Depth × Perplexity (GPT-2)	+0.05	0.28	[−0.04, +0.13]
Pred Depth × Doc Length	+0.03	0.51	[−0.06, +0.11]

Table 4: Predicted depth distribution: Common Crawl 2025-51 vs. balanced lab (C4).

Depth	CC 2025 (%)	Lab C4 (%)
0 (human)	42	16.7
1	31	16.7
2	18	16.7
3+	9	50.0

4.2 Results

Table 3 reports rank correlations between predicted depth and quality metrics. None of the correlations reach practically meaningful effect sizes. The depth–AlignScore correlation is $\rho = -0.08$ ($p = 0.07$), failing to reach statistical significance. Vocabulary richness shows a statistically significant but negligibly small negative correlation ($\rho = -0.14$, $p = 0.002$). Perplexity and document length show no meaningful association with predicted depth.

Table 4 compares the predicted depth distribution on Common Crawl against the balanced lab distribution. The model assigns 42% of Common Crawl documents to depth 0 (human-authored) and only 9% to depth 3+. Given that AI-generated content in recent Common Crawl snapshots is estimated at 15–25%, this suggests substantial false negatives: the model systematically underestimates the prevalence of synthetic text in wild data.

Calibration is poor: the expected calibration error is ECE = 0.31, with mean prediction confidence of only 0.38 ± 0.12. The AUC drops by 0.15 relative to lab conditions, confirming substantial domain shift.

4.3 Analysis

These results expose a severe generalization gap. The 15D features, trained on clean C4 subsets generated by a single model family, fail to capture the distributional complexity of Common Crawl data, which spans multiple languages, formats, and unknown generator pipelines. The negligible depth–AlignScore correlation ($\rho = -0.08$) undermines the claim that K^* boundaries carry actionable significance for data curation: if predicted depth does not track text quality in the wild, filtering by estimated depth is no better than random at scale. The weak vocabulary richness correlation ($\rho = -0.14$), while statistically significant due to large sample size, has an effect size too small ($r^2 < 0.02$) to serve as validation evidence.

The distribution shift (42% depth-0 on Common Crawl vs. 16.7% in lab) further reveals that the classifier's decision boundaries, calibrated on balanced lab data, collapse under real-world class priors and distributional heterogeneity.

Table 5: $K^* = 2$ detection power by sample size (Qwen-1.5B-Inst continuation, C4, $\theta = 0.60$, 10 resamples per n, 10,000 bootstrap iterations).

n per depth	Mean AUC 1v2	SD	Mean CI lower	% K^*=2	% K^*=1
1,000	0.6095	0.028	0.581	40%	60%
2,000	0.6108	0.019	0.592	50%	50%
3,000	0.6115	0.015	0.597	60%	40%
4,000	0.6118	0.012	0.600	80%	20%
5,000	0.6120	0.010	0.602	90%	10%

Table 6: Cross-model power comparison at $n = 3{,}000$ per depth (C4 web text).

Model	AUC 1v2 (full)	Margin over θ	Est. % K^*=2
Pythia-1.4B	0.6539	+5.39pp	95%
Qwen-1.5B base	0.6262	+2.62pp	80%
Qwen-7B	0.6178	+1.78pp	70%
OLMo-1B	0.6145	+1.45pp	65%
LLaMA-3.1-8B	0.6123	+1.23pp	62%
Qwen-1.5B-Inst	0.6112	+1.12pp	60%
Mistral-7B	0.5935	−0.65pp	5%

5 Subsampling Power Analysis (exp-028)

5.1 Setup

We examine how sample size affects the stability of K^* determination. Using the most marginal configuration (Qwen-1.5B-Instruct continuation on C4, full-data AUC 1v2 = 0.6112, CI lower = 0.6015), we construct subsamples at $n \in \{1{,}000,\ 2{,}000,\ 3{,}000,\ 4{,}000,\ 5{,}000\}$ per depth level, with 10 independent resamples per size. For each resample, we compute the bootstrap 95% CI for the 1v2 AUC and determine whether $K^* = 2$ (CI lower $> \theta = 0.60$) or $K^* = 1$.

5.2 Results

Table 5 reports the power curve. At $n = 1{,}000$, only 40% of resamples yield $K^* = 2$; the majority (60%) conclude $K^* = 1$. The detection rate rises monotonically with sample size, reaching 80% at $n = 4{,}000$ and 90% at $n = 5{,}000$. The mean 1v2 AUC is remarkably stable across sample sizes (0.6095–0.6120), confirming that the underlying effect size is genuine but small: the standard deviation shrinks from 0.028 at $n = 1{,}000$ to 0.010 at $n = 5{,}000$, and the CI lower bound rises from 0.581 to 0.602.

Table 6 reveals substantial cross-model heterogeneity in statistical power. At $n = 3{,}000$, Pythia-1.4B (margin +5.39pp over θ) achieves 95% detection rate, while Qwen-cont (+1.12pp) and LLaMA-8B (+1.23pp) reach only 60% and 62% respectively. Mistral-7B, with a negative margin (−0.65pp), achieves only 5% detection at any sample size, consistent with its $K^* = 1$ classification.

5.3 Analysis

The power curve exposes two fundamental issues with discrete K^* characterization.

First, the effect size for the $K^* = 2$ boundary on C4 web text is vanishingly small. The 1v2 AUC of 0.6112 exceeds $\theta = 0.60$ by only 1.12 percentage points (Cohen's $h \approx 0.20$, the lower boundary of "small effect"). Requiring 4,000 samples per depth pair for 80% detection power invites a pointed critique: if a claimed fundamental property of recursive generation requires such large samples to detect, the signal may be too weak to carry practical significance.

Second, the "7/8 configurations yield $K^* = 2$" claim masks substantial power asymmetry across models. Pythia-1.4B needs only $n \approx 1{,}000$ for stable $K^* = 2$ detection, while Qwen-cont and LLaMA-8B require $n \geq 4{,}000$. At any fixed sample size below 4,000, the fraction of models supporting $K^* = 2$ drops, and K^* becomes jointly determined by sample size, threshold, and model identity rather than by the data generating process alone.

When combined with the θ-sensitivity reported in the controlled experiments ($\theta = 0.55$ shifts most models to $K^* = 2{-}3$; $\theta = 0.70$ shifts most to $K^* = 1$), the power analysis reinforces the conclusion that K^* is not a stable property of the text but a function of experimental design parameters $(n, \theta, \text{model})$.

6 Discussion

6.1 The Fragility of Discrete K^*

The controlled experiments (Section 3) present an appealing narrative: recursive generation has a bounded discrimination boundary $K^* \in \{1, 2, 3\}$ with clean domain dependence. The validation experiments (Sections 4–5) dismantle this narrative along two axes.

The Common Crawl evaluation (exp-014) shows that the lab-trained depth estimator does not transfer to wild data. ECE of 0.31 indicates that predicted depths are poorly calibrated, and the negligible correlation with quality metrics ($|\rho| \leq 0.14$ for all proxies) means that K^* boundaries, even if correctly estimated, do not track the quality degradation that motivates depth-aware curation. The practical implication is stark: a curation system built on $K^* = 2$ filtering would not meaningfully improve data quality on Common Crawl-scale corpora.

The power analysis (exp-028) reveals that $K^* = 2$ is a statistically fragile determination. For marginal configurations (Qwen-cont, LLaMA-8B, OLMo-1B), the 1v2 AUC exceeds θ by 1–2 percentage points. Whether these configurations are classified as $K^* = 1$ or $K^* = 2$ depends on sample size and random variation. The "7/8 $K^* = 2$" headline is achievable only at the specific $(n = 5{,}000, \theta = 0.60)$ operating point.

6.2 From Discrete Boundaries to Continuous Decay

These results motivate reframing recursive generation contamination not as a problem of identifying discrete K^* boundaries but as characterizing continuous information decay curves. The DPI guarantee ensures monotonic decay of mutual information with depth, but the decay *rate* varies continuously across models and domains. The 1v2 AUC values across models (0.5935–0.6539 on C4; see Table 6) form a continuum, not discrete clusters. Discretizing this continuum into K^* via a threshold θ discards information about the decay rate and introduces dependence on (n, θ).

A continuous characterization would estimate a decay function $\alpha(k; \text{model}, \text{domain})$ mapping depth k to discriminability. This function captures both the rate of information loss (fast for Wikipedia, slow for arXiv) and the model-specific marginal differences that discrete K^* collapses. For data curation, a continuous score avoids the binary filtering granularity that K^* imposes and enables graduated weighting schemes calibrated to domain-specific decay profiles.

6.3 Limitations

Several limitations constrain interpretation. (1) The controlled experiments use models up to 8B parameters; behavior at 70B+ scale remains untested, though DPI guarantees finite K^*_{theory} regardless of scale. (2) The Common Crawl evaluation uses a single snapshot (2025-51); temporal variation in AI content prevalence may affect distributional shift severity. (3) The power analysis uses a single marginal model (Qwen-cont) as the test case; power characteristics of other marginal

models (OLMo, LLaMA) may differ. (4) We do not evaluate continuous decay curve estimation, which remains a direction for future work.

7 Conclusion

We presented the first systematic measurement of the generational discrimination boundary K^* across eight model configurations, three domains, and four decoding strategies. The controlled experiments yield $K^* \in \{1, 2, 3\}$ with domain-dependent structure: Wikipedia $K^* = 1$, C4 web text $K^* = 2$ (7/8 configurations), arXiv $K^* = 3$ (5/5 domain-competent models).

However, our validation experiments reveal that this discrete characterization is fragile. Naturalistic evaluation on Common Crawl shows severe lab-to-wild transfer degradation (ECE = 0.31, depth–quality $\rho = -0.08$, $p = 0.07$), indicating that K^* boundaries trained on clean subsets do not carry actionable significance for in-the-wild data curation. The subsampling power analysis demonstrates that $K^* = 2$ determination requires $n \geq 4{,}000$ for 80% power, exposing an effect size at the lower limit of detectability. Combined with θ-sensitivity and cross-model power heterogeneity, K^* appears to be a joint function of $(n, \theta, \text{model})$ rather than an intrinsic property of recursive generation.

These findings point toward a continuous information decay framework as a more principled alternative to discrete K^* classification. The rapid saturation of discriminability (within 1–3 passes) confirms that recursive contamination erases generational structure quickly, but the rate of erasure varies continuously across models and domains in ways that discrete K^* fails to capture.

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